Lecture 19

3/4/26
JRA

Housekeeping

- Midterm will be take home
  - Hand out hard copy on Wednesday, March 18th
  - Due by Friday, April 3rd
- Current homework due Friday, March 6th
- Problem session tomorrow 4-6 pm

Specialized Fluorescence Techniques

Fluorescence, as mentioned earlier, is currently the most popular microscopy method used in biology, by far

We have looked at standard fluorescence microscopy modes

- Confocal microscopy
- Widefield fluorescence microscopy w/ deconvolution

Turn now to five specialized fluorescence techniques, each w/ specific strengths & weaknesses

2PFM

(1) Two-Photon Fluorescence Microscopy

This technique is ideally suited for "deep" imaging

Here, "deep" is ~ 2 mm

Technique used to image cortex of living mouse brain

TIRFM

(2) Total Internal Reflection Fluorescence Microscopy

This technique is ideally suited for surface imaging

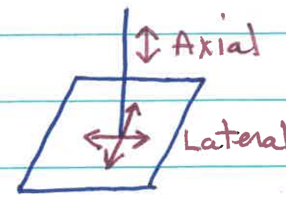
Technique used to image cell membrane & study exocytosis & endocytosis

TIRFM is an axial superresolution technique

LSCM	Axial resolution ≈ 500 nm	Axial = direction of light propagation
TIRFM	Axial resolution ≈ 100 nm	

Lateral resolution is worse than axial resolution in LSCM, typically $\hat{\approx}$ 200 nm. This same lateral resolution applies to TIRFM

The final three techniques we'll study are full superresolution techniques — they beat the diffraction limit in all 3 directions (lateral & axial)



SIM (3) Structured Illumination Microscopy

STED (4) Stimulated Emission Depletion Microscopy

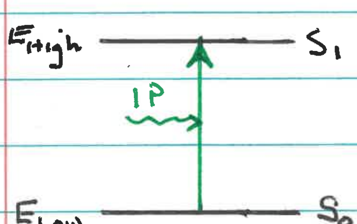
PALM (5) Photoactivated Localization Microscopy

Now go thru each technique in detail (for next several lectures)

Two-Photon Fluorescence Microscopy

This technique is based on two-photon excitation of fluorescence, instead of the usual single-photon excitation

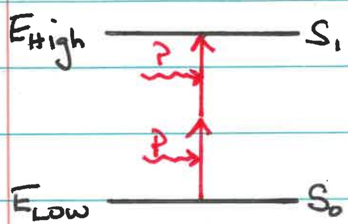
Single-Photon Excitation



In standard single-photon excitation, a fluorophore is excited by a single photon w/ energy corresponding to the energy difference between the states

$$E_{1P\,photon} = E_{High} - E_{Low} = h\nu = h\frac{c}{\lambda}$$

Two-Photon Excitation



In two-photon excitation, a fluorophore is excited by two photons w/ combined energy corresponding to the energy difference between the states

$$E_{2P\ photon} = \frac{1}{2} E_{1P\ photon}$$

This phenomenon was predicted theoretically in the 1930s by Maria Goeppert Mayer (but not confirmed experimentally until the advent of lasers).

Important Attributes of Two-Photon Excitation

(1) 2P Excitation Spectrum

Each 2P photon has ½ the required energy (i.e., ½ the energy of the 1P photon for the same fluorophore)

$\frac{1}{2}$ Energy

2x Wavelength because $E \propto 1/\lambda$

Each 2P photon has twice the wavelength of the 1P photon

Lower E $\Rightarrow$ longer λ

Example) $\lambda_{1P} = 500$ nm — 1 photon

$\lambda_{2P} = 1000$ nm — 2 photon

Two-photon excitation photons typically have wavelengths in the infrared.

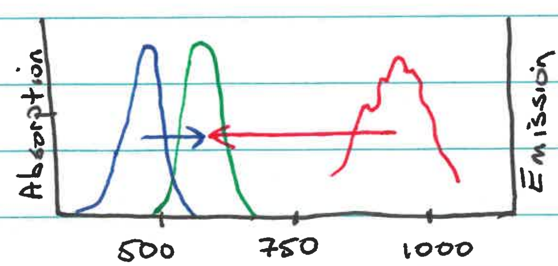
The two-photon excitation spectrum is RED-shifted (i.e., shifted to longer wavelengths) relative to the single-photon excitation spectrum

(2) 2P Emission Spectrum

The two-photon emission spectrum is the same as the single-photon emission spectrum.

Once a fluorophore is in its excited state, it forgets how it got there & decays the same way (via the same emission spectrum) whether excited w/ one or two photons

Example) Looked at 1P & 2P excitation & emission spectra for EGFP (see PP slides)



BLUE — 1P Excitation
RED — 2P Excitation
GREEN — Emission

1P	BLUE Ex → GREEN Em	λ_{Ex} shorter	$\lambda_{Ex} < \lambda_{Em}$
2P	RED Ex → GREEN Em	λ_{Ex} longer	$\lambda_{Ex} > \lambda_{Em}$

Key similarity: The excitation & emission bands are distinct whether it's w/ 1P or 2P excitation, meaning they can be separated w/ suitable filters in each case.

(3) 2P Timing Considerations

The two photons in 2P excitation must arrive at the fluorophore & be absorbed essentially simultaneously (within ~ 10^{-18} sec). Otherwise, there's NO absorption.

The simultaneity requirement can only be satisfied w/ very high light intensities.

SIMULATION Mol Exp Jablonski Diagram

Looked at how delay between photon arrivals affects whether two-photon absorption occurs

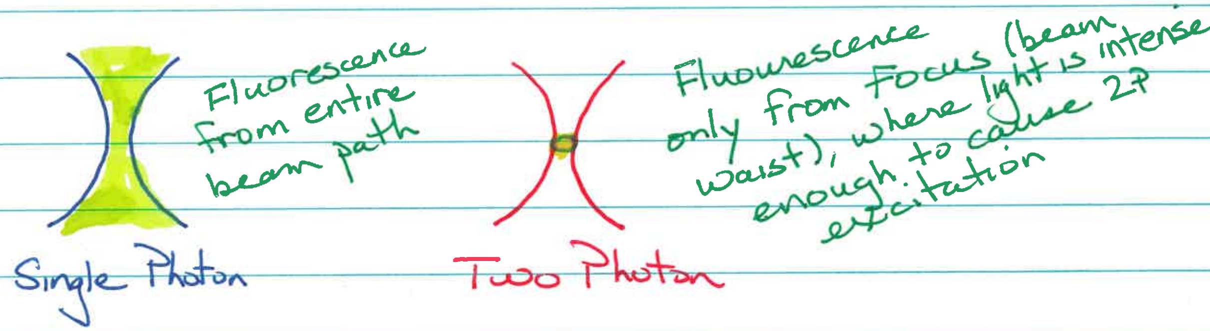
Saw that two photons need to arrive within about 10^{-18} sec of one another to be absorbed

2P Microscopy is a laser scanning technique, like LSCM

Advantages - What are the advantages of 2P microscopy relative to LSCM?

(1) No Out-of-Focus Fluorescence

2P excitation requires very high light intensities. These intensities are only achieved at focus, where the beam is narrowest



• Single Photon (LSCM)

Fluorescence is excited everywhere laser light hits
Need pinhole to exclude out-of-focus fluorescence

• Two Photon

Fluorescence is only excited at focus, where laser light is brightest
Do NOT need pinhole, because no out-of-focus fluorescence is created

(2) Limited Photobleaching / Phototoxicity

The limited photobleaching & phototoxicity caused by 2P excitation is another consequence of the limited excitation volume

- Single Photon (LSCM)

Fluorescence is excited everywhere along light path, so photobleaching & phototoxicity occurs everywhere

- Two Photon

Fluorescence is only excited at focus, so there is very limited photobleaching & phototoxicity

(3) Reduced Scattering

The unique excitation volume in 2P microscopy avoids scattering problems present w/ 1P microscopy, which allows imaging deeper into samples

- Single Photon (LSCM)

Imaging deep into samples requires illuminating & so exciting fluorescence from a considerable sample thickness

Scattering from throughout the light path can cause out-of-focus photons to pass thru the confocal pinhole & contaminate the signal

Scattering can also cause photons from the focal plane to miss the pinhole & so NOT be counted. Deep imaging means photons that should be counted must pass thru more sample to get to the objective & so are more likely to be scattered

Scattering kills LSCM at depths > ~50 μm

- Two Photon

All photons originate in focal plane
No pinhole

Any photons that hit detector necessarily come from the focal plane, so scattering cannot introduce photons from out-of-focus planes

Two photon imaging can be performed at depths up to ~200 μm
